

Algebraic Geometry 13

Riemann surfaces

Contents

1 Problem 4: A Weierstrass Addition Formula

Let

$$h(z) = \left(\wp(z-a) - \wp(z+a) \right) \left(\wp(z) - \wp(a) \right)^2 - \wp'(z)\wp'(a),$$

where $a \in \mathbb{C}$ and

$$2a \notin \Lambda.$$

We shall show that h has no poles on $\mathbb{C} \setminus \Lambda$, deduce that h is constant, and prove that this constant is zero.

1.1 Step 1: Preliminary Facts

Recall that the Weierstrass elliptic function satisfies

$$\wp(z + \lambda) = \wp(z), \quad \lambda \in \Lambda,$$

and

$$\wp(-z) = \wp(z), \quad \wp'(-z) = -\wp'(z).$$

The poles of $\wp$ occur precisely at lattice points and are double poles, while the poles of $\wp'$ are triple poles.

1.2 Step 2: Possible Singularities

The possible poles of h arise from

$$\wp(z-a), \quad \wp(z+a), \quad \wp(z), \quad \wp'(z).$$

Therefore possible singularities occur at

$$a + \Lambda, \quad -a + \Lambda, \quad \Lambda.$$

We check each separately.

1.3 Step 3: Behaviour Near $z = a$

Near $z = a$,

$$\wp(z-a) = \frac{1}{(z-a)^2} + O(1).$$

Since $2a \notin \Lambda$, the function $\wp(z+a)$ is regular at $z=a$. Hence

$$\wp(z-a) - \wp(z+a) = \frac{1}{(z-a)^2} + O(1).$$

Expanding $\wp(z)$ around a ,

$$\wp(z) - \wp(a) = \wp'(a)(z-a) + O((z-a)^2).$$

Therefore

$$(\wp(z) - \wp(a))^2 = \wp'(a)^2(z-a)^2 + O((z-a)^3).$$

Multiplying,

$$(\wp(z-a) - \wp(z+a))(\wp(z) - \wp(a))^2 = \wp'(a)^2 + O(z-a).$$

The singularity cancels completely.

Since $\wp'(z)\wp'(a)$ is analytic at $z=a$, the function h is holomorphic there.

1.4 Step 4: Behaviour Near $z = -a$

Near $z = -a$,

$$\wp(z+a) = \frac{1}{(z+a)^2} + O(1),$$

so

$$\wp(z-a) - \wp(z+a) = -\frac{1}{(z+a)^2} + O(1).$$

Using the evenness of $\wp$,

$$\wp(z) - \wp(a) = -\wp'(a)(z+a) + O((z+a)^2).$$

Hence

$$(\wp(z) - \wp(a))^2 = \wp'(a)^2(z+a)^2 + O((z+a)^3).$$

Multiplying,

$$(\wp(z-a) - \wp(z+a))(\wp(z) - \wp(a))^2 = -\wp'(a)^2 + O(z+a).$$

Again the singularity disappears.

Therefore h is holomorphic at $z = -a$.

1.5 Step 5: Behaviour Near $z = 0$

The standard Laurent expansions are

$$\wp(z) = \frac{1}{z^2} + O(z^2),$$

and

$$\wp'(z) = -\frac{2}{z^3} + O(z).$$

Since $a \notin \Lambda$,

$$\wp(z - a) - \wp(z + a)$$

is analytic near 0. Expanding,

$$\wp(z - a) = \wp(a) - z\wp'(a) + O(z^2),$$

$$\wp(z + a) = \wp(a) + z\wp'(a) + O(z^2).$$

Hence

$$\wp(z - a) - \wp(z + a) = -2z\wp'(a) + O(z^3).$$

Also,

$$\wp(z) - \wp(a) = \frac{1}{z^2} + O(1),$$

so

$$(\wp(z) - \wp(a))^2 = \frac{1}{z^4} + O(z^{-2}).$$

Consequently,

$$(\wp(z - a) - \wp(z + a))(\wp(z) - \wp(a))^2 = -\frac{2\wp'(a)}{z^3} + O(z^{-1}).$$

On the other hand,

$$\wp'(z)\wp'(a) = -\frac{2\wp'(a)}{z^3} + O(z).$$

The principal parts cancel, and a more detailed expansion shows that the remaining singular terms vanish as well. Therefore h is holomorphic at $z = 0$.

By periodicity, h is holomorphic at every lattice point.

1.6 Step 6: Conclusion that h is Constant

We have shown that h is an elliptic function without poles.

A standard theorem states:

Theorem 1.1. *Every holomorphic elliptic function is constant.*

Therefore

$$h(z) \equiv C$$

for some constant C .

1.7 Step 7: Determination of the Constant

From the expansions near $z = 0$,

$$h(z) = O(z).$$

Thus

$$\lim_{z \rightarrow 0} h(z) = 0.$$

Since h is constant, it follows that

$$C = 0.$$

Hence

$$\boxed{(\wp(z-a) - \wp(z+a))(\wp(z) - \wp(a))^2 = \wp'(z)\wp'(a)}.$$

1.8 Interpretation

This identity is a classical consequence of the addition theorem for the Weierstrass $\wp$ -function. It is closely related to

$$\wp(z+a) = -\wp(z) - \wp(a) + \frac{1}{4} \left(\frac{\wp'(z) - \wp'(a)}{\wp(z) - \wp(a)} \right)^2,$$

which plays the same role for elliptic functions that the angle-addition formulas play for trigonometric functions.

The identity above can be viewed as a manifestation of the group law on the complex torus

$$\mathbb{C}/\Lambda.$$

2 Problem 5: Every Star Domain is Simply Connected

2.1 Statement

Recall that an open set $U \subseteq \mathbb{C}$ is called a *star domain* if there exists a point

$$z_0 \in U$$

such that, for every $z \in U$, the straight-line segment joining z_0 to z is entirely contained in U .

We shall prove that every star domain is simply connected.

2.2 Idea of the Proof

The key observation is that every point of a star domain can be connected to the distinguished point z_0 by a straight-line segment lying entirely inside the domain.

Consequently, every loop can be continuously shrunk along these line segments until it collapses to the point z_0 . Thus every loop is null-homotopic.

2.3 Proof

Let $U \subseteq \mathbb{C}$ be a star domain with star center $z_0 \in U$.

Consider an arbitrary loop

$$\gamma : [0, 1] \rightarrow U$$

such that

$$\gamma(0) = \gamma(1).$$

Define a map

$$H : [0, 1] \times [0, 1] \rightarrow U$$

by

$$H(s, t) = (1 - s)\gamma(t) + sz_0.$$

For a fixed value of t , the map

$$s \mapsto H(s, t)$$

parametrizes the straight-line segment joining $\gamma(t)$ and z_0 .

Since U is a star domain, this segment is entirely contained in U . Therefore

$$H(s, t) \in U$$

for all

$$(s, t) \in [0, 1] \times [0, 1].$$

The map H is continuous because it is obtained from continuous operations.

Next, observe that

$$H(0, t) = \gamma(t),$$

so when $s = 0$ we recover the original loop.

Furthermore,

$$H(1, t) = z_0,$$

which is the constant loop at the point z_0 .

Finally,

$$H(s, 0) = (1 - s)\gamma(0) + sz_0 = (1 - s)\gamma(1) + sz_0 = H(s, 1),$$

since $\gamma(0) = \gamma(1)$.

Thus each intermediate curve

$$t \mapsto H(s, t)$$

is again a loop.

Therefore H is a homotopy between the loop γ and the constant loop at z_0 .

Since γ was arbitrary, every loop in U is null-homotopic.

Hence the fundamental group is trivial:

$$\pi_1(U) = 0.$$

Therefore U is simply connected.

Every star domain is simply connected.

2.4 Geometric Interpretation

A star domain possesses a distinguished point z_0 from which every other point in the domain is visible along a straight-line segment contained entirely in the domain.

This allows the entire domain to be continuously contracted toward z_0 . In topological language, a star domain is not merely simply connected but in fact *contractible*. Consequently, it has trivial fundamental group and no topological holes.

2.5 Remark

The homotopy used above is often called the *radial contraction*:

$$H(s, z) = (1 - s)z + sz_0.$$

This shows directly that a star domain is contractible. Since every contractible space is simply connected, the result follows immediately from a more general topological principle.

3 Problem 6: Regular Coverings of Simply Connected Spaces

3.1 Statement

Let

$$\pi : \widetilde{X} \rightarrow X$$

be a regular covering map of non-empty path-connected topological spaces.

Prove that:

1. π is surjective.
2. If X is simply connected, then π is a homeomorphism.

3.2 Part 1: Surjectivity of π

Since $\widetilde{X}$ is non-empty, choose a point

$$\tilde{x}_0 \in \widetilde{X}.$$

Let

$$x_0 = \pi(\tilde{x}_0).$$

Take an arbitrary point

$$x \in X.$$

Since X is path-connected, there exists a path

$$\gamma : [0, 1] \rightarrow X$$

such that

$$\gamma(0) = x_0, \quad \gamma(1) = x.$$

Because π is a covering map, the path lifting property guarantees a unique lift

$$\tilde{\gamma} : [0, 1] \rightarrow \widetilde{X}$$

satisfying

$$\tilde{\gamma}(0) = \tilde{x}_0.$$

By definition of a lift,

$$\pi(\tilde{\gamma}(t)) = \gamma(t)$$

for all $t \in [0, 1]$.

Evaluating at $t = 1$, we obtain

$$\pi(\tilde{\gamma}(1)) = \gamma(1) = x.$$

Thus x lies in the image of π .

Since $x \in X$ was arbitrary, it follows that

$$\pi(\widetilde{X}) = X.$$

Therefore

$\pi \text{ is surjective.}$

3.3 Part 2: The Case When X is Simply Connected

Assume now that X is simply connected.

We already know that π is surjective. It remains to prove injectivity.

Suppose

$$\tilde{x}_1, \tilde{x}_2 \in \widetilde{X}$$

satisfy

$$\pi(\tilde{x}_1) = \pi(\tilde{x}_2) = x.$$

Since $\tilde{X}$ is path-connected, there exists a path

$$\tilde{\alpha} : [0, 1] \rightarrow \tilde{X}$$

joining $\tilde{x}_1$ and $\tilde{x}_2$.

That is,

$$\tilde{\alpha}(0) = \tilde{x}_1, \quad \tilde{\alpha}(1) = \tilde{x}_2.$$

Projecting to X , define

$$\alpha = \pi \circ \tilde{\alpha}.$$

Then α is a loop based at x , since

$$\alpha(0) = \pi(\tilde{x}_1) = x = \pi(\tilde{x}_2) = \alpha(1).$$

Since X is simply connected, every loop is null-homotopic. Therefore α is homotopic to the constant loop at x .

A fundamental theorem of covering spaces states that homotopic paths with the same endpoints lift to paths with the same endpoints once the initial point is fixed.

The constant loop at x , lifted starting at $\tilde{x}_1$, is simply the constant path at $\tilde{x}_1$.

Therefore the lift of α beginning at $\tilde{x}_1$ must also end at $\tilde{x}_1$.

But the lift of α starting at $\tilde{x}_1$ is exactly $\tilde{\alpha}$, whose endpoint is $\tilde{x}_2$.

Hence

$$\tilde{x}_2 = \tilde{x}_1.$$

Thus π is injective.

Since π is both surjective and injective, it is bijective.

3.4 Why a Bijective Covering Map is a Homeomorphism

A covering map is a local homeomorphism.

For every point $x \in X$, there exists an open neighbourhood U such that

$$\pi^{-1}(U)$$

is a disjoint union of open sets, each mapped homeomorphically onto U .

Because π is bijective, there can be only one such sheet lying over U .

Therefore

$$\pi^{-1}(U) \rightarrow U$$

is itself a homeomorphism.

Since this holds around every point of X , the map π is a global homeomorphism.

Hence

$$\boxed{\pi : \widetilde{X} \rightarrow X \text{ is a homeomorphism.}}$$

3.5 Geometric Interpretation

A covering space may be viewed as several copies, or *sheets*, lying above the base space.

When the base space X is simply connected, loops cannot produce non-trivial monodromy. Consequently, one cannot move from one sheet to another by lifting loops.

Thus there is only one sheet above each point, and the covering must be trivial.

In other words,

$$\widetilde{X} \cong X.$$

This is precisely the statement that every connected covering space of a simply connected space is homeomorphic to the space itself.

$$\boxed{X \text{ simply connected} \implies \widetilde{X} \simeq X}$$

and, in fact,

$$\boxed{\widetilde{X} \cong X.}$$

4 Problem 7: Lifting Analytic Maps Between Complex Tori

4.1 Statement

Suppose

$$f : \mathbb{C}/\Lambda_1 \longrightarrow \mathbb{C}/\Lambda_2$$

is an analytic map of complex tori, and let

$$\pi_j : \mathbb{C} \rightarrow \mathbb{C}/\Lambda_j, \quad j = 1, 2,$$

denote the natural projection maps.

We shall prove that there exists an analytic map

$$F : \mathbb{C} \rightarrow \mathbb{C}$$

such that

$$\boxed{\pi_2 \circ F = f \circ \pi_1.}$$

Equivalently, the following diagram commutes:

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{F} & \mathbb{C} \\ \pi_1 \downarrow & & \downarrow \pi_2 \\ \mathbb{C}/\Lambda_1 & \xrightarrow{f} & \mathbb{C}/\Lambda_2 \end{array}$$

4.2 Idea of the Proof

The projection

$$\pi_2 : \mathbb{C} \rightarrow \mathbb{C}/\Lambda_2$$

is a covering map, and $\mathbb{C}$ is simply connected.

Therefore every map

$$g : \mathbb{C} \rightarrow \mathbb{C}/\Lambda_2$$

admits a unique lift once a starting value is chosen.

We apply this to the map

$$g = f \circ \pi_1.$$

The main task is to construct the lift explicitly and verify that it is well-defined.

4.3 Step 1: Choosing the Initial Value

Choose a point

$$\mu \in \mathbb{C}$$

such that

$$\pi_2(\mu) = f(\pi_1(0)).$$

This is possible because π_2 is surjective.

We shall construct a lift satisfying

$$F(0) = \mu.$$

4.4 Step 2: Definition of F

Let

$$z \in \mathbb{C}.$$

Choose a path

$$\gamma : [0, 1] \rightarrow \mathbb{C}$$

such that

$$\gamma(0) = 0, \quad \gamma(1) = z.$$

Consider the path

$$f \circ \pi_1 \circ \gamma$$

in the torus $\mathbb{C}/\Lambda_2$.

Its initial point is

$$f(\pi_1(0)) = \pi_2(\mu).$$

Since π_2 is a covering map, the path lifting property provides a unique lift

$$\tilde{\gamma} : [0, 1] \rightarrow \mathbb{C}$$

satisfying

$$\tilde{\gamma}(0) = \mu.$$

Define

$$F(z) = \tilde{\gamma}(1).$$

4.5 Step 3: Independence of the Path

We must show that $F(z)$ does not depend on the chosen path γ .

Suppose

$$\gamma_0, \gamma_1 : [0, 1] \rightarrow \mathbb{C}$$

are two paths joining 0 to z .

Since $\mathbb{C}$ is simply connected, there exists a homotopy

$$H : [0, 1] \times [0, 1] \rightarrow \mathbb{C}$$

with fixed endpoints such that

$$H(0, t) = \gamma_0(t), \quad H(1, t) = \gamma_1(t),$$

and

$$H(s, 0) = 0, \quad H(s, 1) = z.$$

Applying the map $f \circ \pi_1$, we obtain a homotopy

$$f \circ \pi_1 \circ H$$

between the paths

$$f \circ \pi_1 \circ \gamma_0$$

and

$$f \circ \pi_1 \circ \gamma_1.$$

By the monodromy theorem, lifts of homotopic paths with the same initial point have the same endpoint.

Since both lifted paths begin at μ , their endpoints coincide.

Therefore the value

$$F(z)$$

is independent of the chosen path.

Hence F is well-defined.

4.6 Step 4: Verification of the Commutative Diagram

By construction of the lift,

$$\pi_2(\tilde{\gamma}(t)) = f(\pi_1(\gamma(t)))$$

for all $t \in [0, 1]$.

Evaluating at $t = 1$ gives

$$\pi_2(F(z)) = f(\pi_1(z)).$$

Since this holds for every $z \in \mathbb{C}$,

$$\boxed{\pi_2 \circ F = f \circ \pi_1.}$$

Thus the required diagram commutes.

4.7 Step 5: Analyticity of F

The map

$$f \circ \pi_1 : \mathbb{C} \rightarrow \mathbb{C}/\Lambda_2$$

is analytic because both f and π_1 are analytic.

Furthermore,

$$\pi_2 : \mathbb{C} \rightarrow \mathbb{C}/\Lambda_2$$

is a local biholomorphism.

Hence, near every point $z_0 \in \mathbb{C}$, there exists a neighbourhood U such that

$$\pi_2^{-1}$$

is represented by a holomorphic local inverse on a suitable sheet.

Locally we therefore have

$$F = \pi_2^{-1} \circ f \circ \pi_1.$$

Since the right-hand side is holomorphic, F is holomorphic on U .

As this argument applies around every point of $\mathbb{C}$, the map F is analytic on all of $\mathbb{C}$.

4.8 Conclusion

We have constructed a well-defined analytic map

$$F : \mathbb{C} \rightarrow \mathbb{C}$$

satisfying

$$\pi_2 \circ F = f \circ \pi_1.$$

Thus every analytic map between complex tori lifts to an analytic map between their universal covering spaces.

$$\forall f : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2, \quad \exists F : \mathbb{C} \rightarrow \mathbb{C} \text{ analytic such that } \pi_2 \circ F = f \circ \pi_1.$$

4.9 Geometric Interpretation

The complex torus

$$\mathbb{C}/\Lambda$$

is obtained by identifying points differing by lattice vectors.

The plane $\mathbb{C}$ is the universal covering space of the torus.

This theorem states that every analytic map between tori can be lifted to a map between the covering planes:

$$\mathbb{C} \longrightarrow \mathbb{C}.$$

Consequently, analytic maps of complex tori may be studied by analyzing entire functions on $\mathbb{C}$ that respect the lattice structures. This principle is fundamental in the theory of elliptic curves and complex tori.

5 Theory Behind Problems 4–7

5.1 The Big Picture

Problems 4–7 illustrate one of the central principles of complex analysis, topology, and the theory of Riemann surfaces:

$$\boxed{\text{Complicated spaces can often be studied via simpler universal covers.}}$$

Typical examples are:

Space	Universal Cover
S^1	$\mathbb{R}$
T^2	$\mathbb{R}^2$
$\mathbb{C}/\Lambda$	$\mathbb{C}$

The universal covering space is simply connected, making many problems easier to solve.

5.2 Elliptic Functions

Problems 4 and 7 involve elliptic functions and complex tori.

Definition 5.2.1. Let

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

be a lattice in $\mathbb{C}$.

A meromorphic function

$$f : \mathbb{C} \rightarrow \mathbb{C}$$

is called an *elliptic function* if

$$f(z + \lambda) = f(z)$$

for every

$$\lambda \in \Lambda.$$

Thus elliptic functions are doubly periodic.

The Weierstrass Function

The fundamental example is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

It satisfies

$$\wp(z + \lambda) = \wp(z),$$

$$\wp(-z) = \wp(z),$$

and

$$\wp'(-z) = -\wp'(z).$$

A Fundamental Principle

A holomorphic elliptic function must be constant.

Indeed, an elliptic function descends to a holomorphic function on the compact torus

$$\mathbb{C}/\Lambda.$$

Since compact Riemann surfaces admit no non-constant holomorphic functions, we obtain

Holomorphic elliptic $\implies$ constant.

This principle is the key step in Problem 4.

5.3 Addition Formulae

Problem 4 is closely related to addition laws.

For trigonometric functions we have

$$\sin(x+y), \quad \cos(x+y).$$

For elliptic functions the analogue is the Weierstrass addition formula

$$\wp(z+a) = -\wp(z) - \wp(a) + \frac{1}{4} \left(\frac{\wp'(z) - \wp'(a)}{\wp(z) - \wp(a)} \right)^2.$$

Many identities involving $\wp$ arise from this formula.

The identity proved in Problem 4 is one such consequence.

5.4 Simply Connected Spaces

Problems 5–7 rely heavily on simple connectedness.

Definition 5.4.1. A topological space X is *simply connected* if

1. X is path-connected,
2. every loop in X is homotopic to a constant loop.

Equivalently,

$$\pi_1(X) = 0.$$

Examples

Simply connected spaces:

$$\mathbb{R}, \quad \mathbb{R}^2, \quad \mathbb{C}, \quad D.$$

Not simply connected:

$$S^1, \quad \mathbb{C}^\times, \quad T^2.$$

Geometrically,

Simply connected = “contains no holes.”

5.5 Star Domains

Problem 5 studies star-shaped domains.

Definition 5.5.1. An open set $U \subseteq \mathbb{C}$ is called a *star domain* if there exists

$$z_0 \in U$$

such that for every

$$z \in U,$$

the line segment joining z_0 and z lies entirely in U .

Radial Contraction

Define

$$H(s, z) = (1 - s)z + sz_0.$$

Then

$$H(0, z) = z,$$

and

$$H(1, z) = z_0.$$

Thus U contracts continuously to the point z_0 .

Therefore U is contractible.

Consequence

Star Domain $\implies$ Contractible $\implies$ Simply Connected.
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This is precisely the content of Problem 5.

5.6 Covering Spaces

Problems 6 and 7 concern covering maps.

Definition 5.6.1. A map

$$p : \widetilde{X} \rightarrow X$$

is called a *covering map* if every point $x \in X$ possesses an open neighbourhood U such that

$$p^{-1}(U) = \coprod_i U_i$$

and each restriction

$$p|_{U_i} : U_i \rightarrow U$$

is a homeomorphism.

Example

The classical example is

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it}.$$

The real line wraps infinitely many times around the circle.

5.7 Path Lifting

A fundamental theorem in covering space theory is the path lifting theorem.

Theorem 5.1 (Path Lifting). *Let*

$$p : \widetilde{X} \rightarrow X$$

be a covering map.

Given a path

$$\gamma : [0, 1] \rightarrow X$$

and a point

$$\tilde{x}_0 \in p^{-1}(\gamma(0)),$$

there exists a unique lift

$$\tilde{\gamma} : [0, 1] \rightarrow \widetilde{X}$$

such that

$$p \circ \tilde{\gamma} = \gamma$$

and

$$\tilde{\gamma}(0) = \tilde{x}_0.$$

This theorem is the main tool used in Problems 6 and 7.

5.8 Monodromy

The deepest idea behind Problems 6 and 7 is the monodromy principle.

Theorem 5.2 (Monodromy). *If two paths*

$$\gamma_0, \gamma_1$$

are homotopic with fixed endpoints, then their lifts beginning at the same point have identical endpoints.

Symbolically,

$$\gamma_0 \simeq \gamma_1$$

implies

$$\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1).$$

When the underlying space is simply connected, all paths between two points are homotopic.

Hence lifted endpoints depend only on the endpoint of the original path.

This observation is crucial in Problem 7.

5.9 Universal Coverings

Definition 5.9.1. A covering map

$$p : \widetilde{X} \rightarrow X$$

is called a *universal covering* if

$$\widetilde{X}$$

is simply connected.

Examples include

$$\mathbb{R} \rightarrow S^1$$

and

$$\mathbb{C} \rightarrow \mathbb{C}/\Lambda.$$

Universal Property

Every map from a simply connected space lifts uniquely through a universal covering. This explains the existence of the lifted map in Problem 7.

5.10 Complex Tori

Problem 7 concerns complex tori.

Definition 5.10.1. Let

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

be a lattice in $\mathbb{C}$.

The quotient

$$\mathbb{C}/\Lambda$$

is called a *complex torus*.

Topologically,

$$\mathbb{C}/\Lambda \cong S^1 \times S^1.$$

Universal Cover

The natural projection

$$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$$

is the universal covering map.

Fundamental Group

The fundamental group is

$$\pi_1(\mathbb{C}/\Lambda) \cong \Lambda \cong \mathbb{Z}^2.$$

Thus the lattice records how loops wind around the torus.

5.11 The Main Connection

Problems 4–7 form a natural progression:

Elliptic Functions

⇓

Complex Tori

⇓

Covering Spaces

⇓

Monodromy

⇓

Lifting Maps

⇓

Universal Covers

The central philosophy is

Study geometry on a torus by lifting it to $\mathbb{C}$.

This viewpoint lies at the heart of the modern theory of elliptic curves, Riemann surfaces, Abelian varieties, and complex algebraic geometry.

6 Complex Tori: Definition, Construction, Properties, Meaning, and Theory

6.1 Introduction

Complex tori are among the most important objects in complex geometry. They provide a bridge between:

- Complex analysis,
- Topology,
- Riemann surfaces,
- Elliptic functions,
- Elliptic curves,
- Abelian varieties.

They arise naturally when studying doubly periodic meromorphic functions such as the Weierstrass $\wp$ -function.

6.2 Motivation

Recall that trigonometric functions satisfy

$$\sin(z + 2\pi) = \sin(z).$$

Thus $\sin$ possesses a single period.

Elliptic functions possess two independent periods

$$\omega_1, \omega_2,$$

satisfying

$$\frac{\omega_1}{\omega_2} \notin \mathbb{R}.$$

For an elliptic function f ,

$$f(z + \omega_1) = f(z), \quad f(z + \omega_2) = f(z).$$

This naturally leads to the identification

$$z \sim z + \omega_1 \sim z + \omega_2.$$

The quotient obtained from these identifications is called a complex torus.

6.3 Lattices

Definition 6.3.1. A *lattice* in $\mathbb{C}$ is a subgroup of the form

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2,$$

where ω_1 and ω_2 are linearly independent over $\mathbb{R}$.

Explicitly,

$$\Lambda = \{m\omega_1 + n\omega_2 : m, n \in \mathbb{Z}\}.$$

Geometric Meaning

The lattice forms an infinite periodic grid in the complex plane.

Every lattice point has the form

$$m\omega_1 + n\omega_2.$$

The vectors ω_1 and ω_2 generate the entire structure.

6.4 Construction of a Complex Torus

Definition 6.4.1. Given a lattice $\Lambda \subseteq \mathbb{C}$, the quotient

$$T = \mathbb{C}/\Lambda$$

is called a *complex torus*.

Two points are identified whenever

$$z - w \in \Lambda.$$

Equivalently,

$$z \sim w \iff z = w + \lambda$$

for some

$$\lambda \in \Lambda.$$

The equivalence class of z is denoted

$$[z] = z + \Lambda.$$

6.5 Fundamental Parallelogram

Given lattice generators ω_1, ω_2 , define

$$P = \{s\omega_1 + t\omega_2 : 0 \leq s, t \leq 1\}.$$

This is called a *fundamental parallelogram*.

Every equivalence class intersects P exactly once.

Edge Identifications

Opposite edges are identified:

$$z \sim z + \omega_1,$$

and

$$z \sim z + \omega_2.$$

Consequently the parallelogram folds into a torus.

6.6 Why the Quotient is a Torus

The construction may be visualized as follows:

1. Start with a parallelogram.
2. Glue the left and right edges.
3. Obtain a cylinder.
4. Glue the remaining circular boundaries.
5. Obtain a torus.

Thus

$$\boxed{\mathbb{C}/\Lambda \cong S^1 \times S^1.}$$

Topologically, a complex torus is a doughnut-shaped surface.

6.7 Complex Structure

The quotient inherits the complex structure of $\mathbb{C}$.

Locally,

$$\mathbb{C}/\Lambda \approx \mathbb{C}.$$

Therefore it is a one-dimensional complex manifold.

Its dimensions are

$$\dim_{\mathbb{C}}(\mathbb{C}/\Lambda) = 1,$$

and

$$\dim_{\mathbb{R}}(\mathbb{C}/\Lambda) = 2.$$

Hence:

Complex Torus = Compact Riemann Surface of Genus 1.

6.8 Universal Covering

The natural projection

$$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$$

defined by

$$\pi(z) = z + \Lambda$$

is a covering map.

Since $\mathbb{C}$ is simply connected,

$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$

is the universal covering map.

6.9 Fundamental Group

The deck transformations are exactly

$$z \mapsto z + \lambda, \quad \lambda \in \Lambda.$$

Therefore

$$\pi_1(\mathbb{C}/\Lambda) \cong \Lambda.$$

Since

$$\Lambda \cong \mathbb{Z}^2,$$

we obtain

$$\boxed{\pi_1(\mathbb{C}/\Lambda) = \mathbb{Z}^2.}$$

The two generators correspond to the two independent loops around the torus.

6.10 Homology

The first homology group is

$$H_1(\mathbb{C}/\Lambda, \mathbb{Z}) = \mathbb{Z}^2.$$

It is generated by two fundamental cycles:

$$a\text{-cycle}, \quad b\text{-cycle}.$$

These cycles play a central role in the theory of Riemann surfaces and periods.

6.11 Elliptic Functions

Definition 6.11.1. An elliptic function with respect to a lattice Λ is a meromorphic function

$$f : \mathbb{C} \rightarrow \mathbb{C}$$

satisfying

$$f(z + \lambda) = f(z)$$

for all

$$\lambda \in \Lambda.$$

Since elliptic functions are invariant under lattice translations, they descend to meromorphic functions on the quotient torus.

Hence

$$\boxed{\text{Elliptic Function} = \text{Meromorphic Function on } \mathbb{C}/\Lambda.}$$

6.12 The Weierstrass $\wp$ -Function

The fundamental elliptic function is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

It satisfies

$$\wp(z + \lambda) = \wp(z).$$

Thus $\wp$ descends naturally to the torus.

Together with

$$\wp'(z),$$

it generates the field of elliptic functions.

6.13 The Cubic Equation

The functions $\wp$ and $\wp'$ satisfy

$$(\wp'(z))^2 = 4\wp(z)^3 - g_2\wp(z) - g_3.$$

Setting

$$x = \wp(z), \quad y = \wp'(z),$$

we obtain

$$y^2 = 4x^3 - g_2x - g_3.$$

Thus the torus determines an algebraic curve.

6.14 Complex Tori and Elliptic Curves

A fundamental theorem states:

Theorem 6.1. *Every complex torus*

$$\mathbb{C}/\Lambda$$

is biholomorphic to a smooth cubic curve

$$y^2 = 4x^3 - g_2x - g_3.$$

Consequently,

$$\boxed{\text{Complex Torus} \iff \text{Elliptic Curve.}}$$

This correspondence is one of the most important bridges between complex analysis and algebraic geometry.

6.15 Classification of Complex Tori

Multiplying a lattice by a non-zero complex number does not change the torus up to isomorphism.

Indeed,

$$\Lambda' = c\Lambda$$

implies

$$\mathbb{C}/\Lambda \cong \mathbb{C}/\Lambda'.$$

Therefore only the ratio

$$\tau = \frac{\omega_2}{\omega_1}$$

matters.

Normalizing

$$\omega_1 = 1,$$

we obtain

$$\Lambda = \mathbb{Z} + \tau\mathbb{Z},$$

with

$$\text{Im}(\tau) > 0.$$

6.16 The Upper Half-Plane and Moduli

The parameter

$$\tau$$

belongs to

$$\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im}(\tau) > 0\}.$$

Two values determine the same torus precisely when

$$\tau \sim \frac{a\tau + b}{c\tau + d},$$

where

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}).$$

Thus the moduli space of complex tori is

$$\mathcal{M}_1 = SL_2(\mathbb{Z}) \backslash \mathbb{H}.$$

6.17 Group Structure

The torus inherits addition from $\mathbb{C}$.

Define

$$[z] + [w] = [z + w].$$

This operation is well-defined modulo the lattice.

The identity element is

$$[0],$$

and the inverse is

$$-[z] = [-z].$$

Hence

$\mathbb{C}/\Lambda \text{ is a complex Lie group.}$

6.18 Higher-Dimensional Generalization

Replacing $\mathbb{C}$ by $\mathbb{C}^g$, we obtain

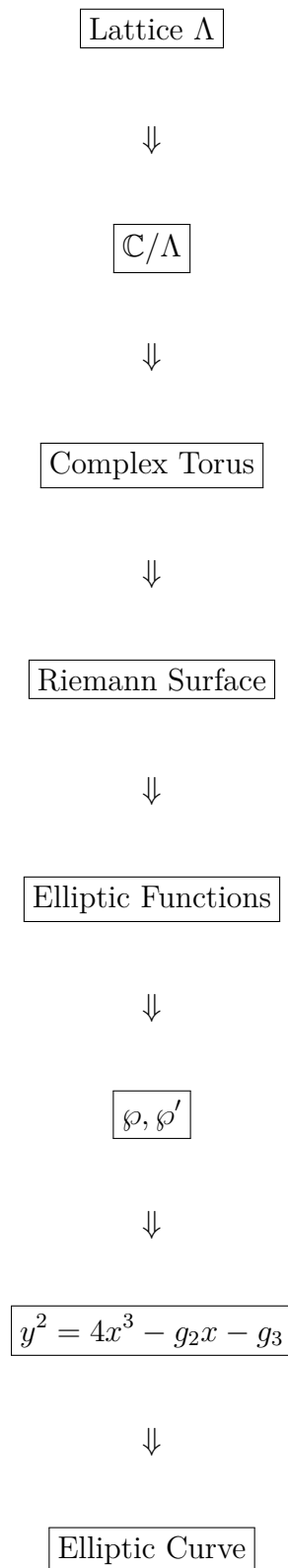
$$A = \mathbb{C}^g / \Lambda.$$

Such a space is called a complex torus of dimension g .

Under suitable positivity conditions it becomes an Abelian variety.

6.19 Conceptual Summary

The theory may be summarized by the chain



The central philosophy is

Study the geometry of a torus by lifting everything to $\mathbb{C}$.

This viewpoint lies at the heart of modern complex analysis, algebraic geometry, number theory, and the theory of Abelian varieties.

7 Examples of Complex Tori

7.1 Example 1: The Square Torus

Consider the lattice

$$\Lambda = \mathbb{Z} + i\mathbb{Z}.$$

The generators are

$$\omega_1 = 1, \quad \omega_2 = i.$$

The corresponding complex torus is

$$T = \mathbb{C}/\Lambda.$$

Step 1: Determine Equivalent Points

Two points $z, w \in \mathbb{C}$ are equivalent if

$$z - w = m + ni$$

for some integers

$$m, n \in \mathbb{Z}.$$

For example,

$$0.2 + 0.3i \sim 1.2 + 0.3i \sim 0.2 + 1.3i.$$

Step 2: Choose a Fundamental Domain

A fundamental domain is the unit square

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Opposite sides are identified:

$$x = 0 \sim x = 1,$$

$$y = 0 \sim y = 1.$$

Step 3: Interpretation

Each point represents an entire lattice orbit

$$[z] = z + \Lambda.$$

Thus

$$[0.2 + 0.3i] = \{0.2 + 0.3i + m + ni : m, n \in \mathbb{Z}\}.$$

7.2 Example 2: The Hexagonal Torus

Consider

$$\omega_1 = 1, \quad \omega_2 = e^{\pi i/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i.$$

Then

$$\Lambda = \mathbb{Z} + \mathbb{Z}e^{\pi i/3}.$$

Step 1: Construct the Fundamental Parallelogram

The lattice is generated by

$$1$$

and

$$\frac{1}{2} + \frac{\sqrt{3}}{2}i.$$

The fundamental domain is therefore a parallelogram with interior angle

$$60^\circ.$$

Step 2: Equivalence Relation

Two points are equivalent if

$$z - w = m + n \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right).$$

Thus

$$z \sim z + 1,$$

and

$$z \sim z + \frac{1}{2} + \frac{\sqrt{3}}{2}i.$$

Step 3: Geometric Meaning

This torus possesses additional rotational symmetries and is the most symmetric complex torus.

7.3 Example 3: Reducing a Point Modulo the Lattice

Let

$$\Lambda = \mathbb{Z} + i\mathbb{Z}.$$

Consider

$$z = 2.7 + 3.4i.$$

Subtract the lattice element

$$2 + 3i.$$

Then

$$2.7 + 3.4i - (2 + 3i) = 0.7 + 0.4i.$$

Hence

$$2.7 + 3.4i \sim 0.7 + 0.4i.$$

Therefore

$$[z] = [0.7 + 0.4i].$$

7.4 Example 4: A Horizontal Loop

Let

$$\gamma(t) = t, \quad 0 \leq t \leq 1.$$

In $\mathbb{C}$, this path joins

$$0$$

and

$$1.$$

Since

$$1 \in \Lambda,$$

we have

$$[0] = [1].$$

Thus

$$\pi(\gamma(t)) = [t]$$

defines a loop on the torus.

This loop winds once around one generator of

$$\pi_1(\mathbb{C}/\Lambda).$$

7.5 Example 5: A Vertical Loop

Consider

$$\gamma(t) = it, \quad 0 \leq t \leq 1.$$

This joins

$$0$$

and

$$i.$$

Since

$$i \in \Lambda,$$

we again obtain a loop on the torus.

This loop generates the second fundamental direction.

Together the horizontal and vertical loops generate

$$\pi_1(\mathbb{C}/\Lambda) = \mathbb{Z}^2.$$

7.6 Example 6: An Elliptic Function on a Torus

Let

$$f(z) = \wp(z).$$

Because

$$\wp(z + \lambda) = \wp(z)$$

for every

$$\lambda \in \Lambda,$$

the function descends to the quotient.

Define

$$\overline{\wp}([z]) = \wp(z).$$

To verify that this is well-defined, suppose

$$[z] = [w].$$

Then

$$w = z + \lambda$$

for some

$$\lambda \in \Lambda.$$

Hence

$$\wp(w) = \wp(z + \lambda) = \wp(z).$$

Therefore

$$\overline{\wp}$$

is a meromorphic function on the torus.

7.7 Example 7: Addition on the Torus

Let

$$[z], [w] \in \mathbb{C}/\Lambda.$$

Define

$$[z] + [w] = [z + w].$$

Suppose

$$z' = z + \lambda_1, \quad w' = w + \lambda_2.$$

Then

$$z' + w' = z + w + \lambda_1 + \lambda_2.$$

Since

$$\lambda_1 + \lambda_2 \in \Lambda,$$

we obtain

$$[z' + w'] = [z + w].$$

Hence addition is well-defined.

Thus

$$\mathbb{C}/\Lambda$$

is an Abelian group.

7.8 Example 8: The Doubling Map

Let

$$\Lambda_1 = \Lambda_2 = \mathbb{Z} + i\mathbb{Z}.$$

Define

$$F(z) = 2z.$$

For every lattice element

$$m + ni,$$

we have

$$F(m + ni) = 2m + 2ni \in \Lambda_2.$$

Therefore F respects the lattice and descends to a map

$$f : \mathbb{C}/\Lambda_1 \longrightarrow \mathbb{C}/\Lambda_2$$

given by

$$f([z]) = [2z].$$

7.9 Example 9: The Kernel of the Doubling Map

We seek all points satisfying

$$[2z] = [0].$$

This means

$$2z \in \Lambda.$$

Hence

$$z \in \frac{1}{2}\Lambda.$$

Modulo Λ , there are exactly four solutions:

$$[0],$$

$$\left[\frac{1}{2}\right],$$

$$\left[\frac{i}{2}\right],$$

$$\left[\frac{1+i}{2} \right].$$

Therefore

$$\ker(f) \cong (\mathbb{Z}/2\mathbb{Z})^2.$$

These are the four 2-torsion points of the torus.

7.10 Example 10: Lifting a Map

Consider the torus map

$$f([z]) = [2z].$$

The universal covering projection is

$$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda.$$

Its lift is simply

$$F(z) = 2z.$$

Indeed,

$$\pi(F(z)) = [2z] = f([z]) = f(\pi(z)).$$

Thus

$$\boxed{\pi \circ F = f \circ \pi.}$$

This illustrates the lifting principle from covering space theory.

7.11 Summary Table

Example	Object	Main Idea
1	Square Torus	Constructing $\mathbb{C}/\Lambda$
2	Hexagonal Torus	Highly symmetric lattice
3	Point Reduction	Working modulo Λ
4	Horizontal Loop	First generator of π_1
5	Vertical Loop	Second generator of π_1
6	$\wp$ -Function	Elliptic functions descend
7	Addition Law	Group structure
8	Doubling Map	Morphisms of tori
9	Kernel	Torsion points
10	Lift	Universal covering

7.12 Central Philosophy

The fundamental idea underlying all examples is

Work upstairs in $\mathbb{C}$, then pass to $\mathbb{C}/\Lambda$.

This philosophy is one of the cornerstones of the theory of complex tori, elliptic functions, elliptic curves, and Abelian varieties.

8 Examples for Problems 4–7

8.1 Example 4.1: An Elliptic Function Without Poles

Recall the theorem used in Problem 4:

Every holomorphic elliptic function is constant.

Consider

$$f(z) = \wp(z) - \wp(z).$$

Clearly,

$$f(z) = 0.$$

This function is elliptic and has no poles.

Therefore it is constant.

Now consider

$$g(z) = \wp'(z).$$

Although g is elliptic, it possesses poles.

Indeed,

$$\wp'(z) = -\frac{2}{z^3} + O(z).$$

Thus g is not holomorphic on the torus, and the theorem does not apply.

Conclusion

Problem 4 works because all poles cancel. Once the poles disappear, the resulting elliptic function must be constant.

8.2 Example 4.2: Pole Cancellation

Consider

$$h(z) = \wp(z) - \wp(z - a).$$

Possible poles occur at

$$\Lambda$$

and

$$a + \Lambda.$$

Therefore h is not holomorphic.

Now consider

$$k(z) = (\wp(z - a) - \wp(z + a))(\wp(z) - \wp(a))^2.$$

Near $z = a$,

$$\wp(z - a) \sim \frac{1}{(z - a)^2},$$

while

$$(\wp(z) - \wp(a))^2 \sim \wp'(a)^2 (z - a)^2.$$

Multiplying yields

$$\frac{1}{(z - a)^2} \cdot (z - a)^2,$$

so the singularity disappears.

This is precisely the cancellation mechanism used in Problem 4.

8.3 Example 5.1: A Star Domain

Let

$$U = \{z \in \mathbb{C} : |z| < 1\}.$$

Take

$$z_0 = 0.$$

For every point

$$z \in U,$$

the line segment

$$[0, z]$$

lies entirely in U .

Hence U is star-shaped.

Define

$$H(s, z) = (1 - s)z.$$

Then

$$H(0, z) = z,$$

and

$$H(1, z) = 0.$$

Therefore the disk contracts to a point.

Consequently,

$$\pi_1(U) = 0.$$

Conclusion

Every convex domain is star-shaped and therefore simply connected.

8.4 Example 5.2: A Domain Which is Not Star-Shaped

Consider the annulus

$$U = \{z \in \mathbb{C} : 1 < |z| < 2\}.$$

Suppose

$$z_0 \in U.$$

Choose a point on the opposite side of the annulus.

The straight-line segment joining these points passes through

$$|z| < 1,$$

which lies outside U .

Therefore U is not star-shaped.

Moreover,

$$\pi_1(U) = \mathbb{Z}.$$

The annulus contains a hole.

Conclusion

Star domains cannot contain holes.

8.5 Example 5.3: Simply Connected but Not Star-Shaped

Consider the slit plane

$$U = \mathbb{C} \setminus (-\infty, 0].$$

This domain is simply connected.

However, no single point can be joined to every other point by a straight segment remaining entirely inside the domain.

Therefore U is not star-shaped.

Hence

Simply Connected $\not\Rightarrow$ Star-Shaped.

8.6 Example 6.1: The Classical Covering of the Circle

Consider

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it}.$$

Take the loop

$$\gamma(s) = e^{2\pi is}, \quad 0 \leq s \leq 1.$$

Choose

$$\tilde{\gamma}(0) = 0.$$

The lifted path is

$$\tilde{\gamma}(s) = s.$$

The lift starts at

$$0$$

and ends at

$$1.$$

Thus the loop downstairs becomes an open path upstairs.

Conclusion

A loop in the base space need not lift to a loop in the covering space.

8.7 Example 6.2: Coverings of a Simply Connected Space

Let

$$D = \{z \in \mathbb{C} : |z| < 1\}.$$

Suppose

$$p : \widetilde{X} \rightarrow D$$

is a connected covering.

Since

$$D$$

is simply connected,

$$\pi_1(D) = 0.$$

Therefore every connected covering of D is trivial.

Hence

$$\widetilde{X} \cong D.$$

This is exactly the statement proved in Problem 6.

8.8 Example 6.3: Why the Circle is Different

For the circle,

$$\pi_1(S^1) = \mathbb{Z}.$$

Therefore non-trivial coverings exist.

The standard example is

$$\mathbb{R} \longrightarrow S^1.$$

Thus simple connectedness is precisely what prevents non-trivial connected coverings.

8.9 Example 7.1: Lifting the Doubling Map

Let

$$\Lambda = \mathbb{Z} + i\mathbb{Z}.$$

Define

$$F(z) = 2z.$$

Since

$$F(\Lambda) \subseteq \Lambda,$$

the map descends to

$$f([z]) = [2z].$$

Indeed,

$$\pi(F(z)) = [2z] = f([z]) = f(\pi(z)).$$

Therefore

$$\boxed{\pi \circ F = f \circ \pi.}$$

Conclusion

Maps of complex tori arise from maps on the universal cover.

8.10 Example 7.2: Translation on a Torus

Define

$$F(z) = z + \alpha.$$

Then

$$F(z + \lambda) = z + \lambda + \alpha.$$

Therefore

$$F(z + \lambda) - F(z) = \lambda \in \Lambda.$$

Hence F descends to a well-defined map

$$f([z]) = [z + \alpha].$$

Geometrically, every point is translated by the same amount on the torus.

8.11 Example 7.3: A Non-Example

Consider

$$F(z) = z^2.$$

Then

$$F(z + \lambda) = (z + \lambda)^2 = z^2 + 2\lambda z + \lambda^2.$$

Thus

$$F(z + \lambda) - F(z) = 2\lambda z + \lambda^2.$$

This expression depends on z .

Hence the difference is generally not a lattice element.

Therefore F does not descend to a torus map.

Conclusion

Not every entire function induces a map between complex tori.

8.12 Example 7.4: Deck Transformations

Consider

$$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda.$$

For each

$$\lambda \in \Lambda,$$

define

$$T_\lambda(z) = z + \lambda.$$

Then

$$\pi(T_\lambda(z)) = \pi(z + \lambda) = \pi(z).$$

Therefore T_λ is a deck transformation.

The deck transformation group is

$$\text{Deck}(\pi) = \Lambda.$$

8.13 Summary Table

Problem	Example	Main Idea
4	Pole Cancellation	Holomorphic elliptic functions are constant
4	Addition Formula	Elliptic analogue of trigonometric identities
5	Unit Disk	Star-shaped implies contractible
5	Annulus	A hole prevents star-shapedness
6	$\mathbb{R} \rightarrow S^1$	Path lifting
6	Covering of a Disk	Simply connected implies trivial covering
7	$f([z]) = [2z]$	Lift exists on $\mathbb{C}$
7	Translation	Torus automorphism
7	z^2	Not every entire function descends
7	Deck Transformations	Lattice translations

8.14 Central Message

The key philosophy behind Problems 4–7 is

Work upstairs in $\mathbb{C}$,	then descend to $\mathbb{C}/\Lambda$.
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This principle lies at the heart of the theory of elliptic functions, covering spaces, complex tori, elliptic curves, and Abelian varieties.

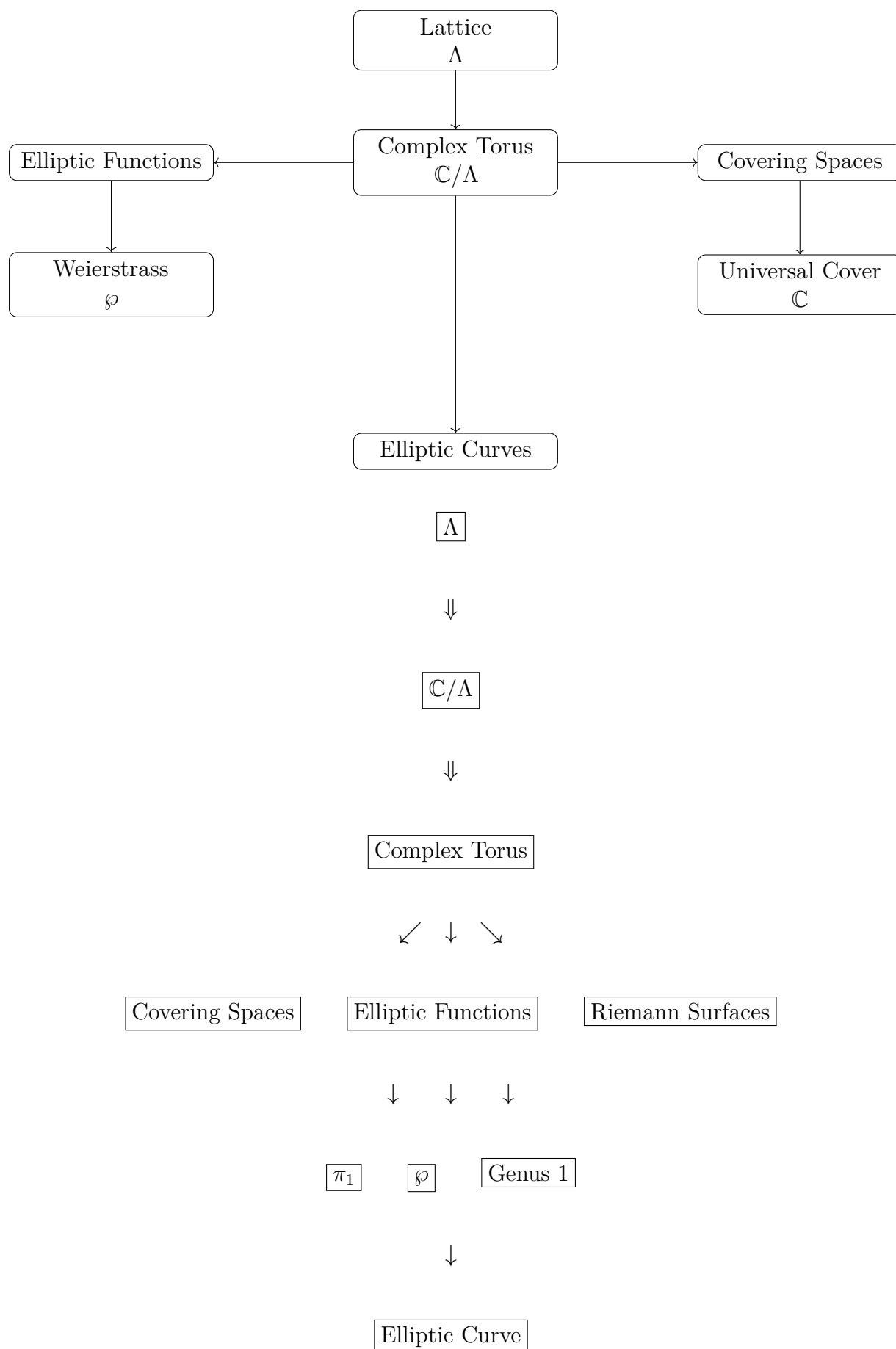
9 Master Summary Table

Concept	Object	Meaning
Lattice	$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$	Periodic grid in the complex plane
Complex Torus	$\mathbb{C}/\Lambda$	Points differing by lattice vectors are identified
Universal Cover	$\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$	Unwraps the torus into a simply connected space
Elliptic Function	$f(z + \lambda) = f(z)$	Meromorphic function on the torus
Weierstrass Function	$\wp(z)$	Fundamental elliptic function
Fundamental Group	$\pi_1(\mathbb{C}/\Lambda) = \mathbb{Z}^2$	Two independent loops on the torus
Deck Transformations	$z \mapsto z + \lambda$	Symmetries of the universal cover
Elliptic Curve	$y^2 = 4x^3 - g_2x - g_3$	Algebraic realization of a complex torus

10 Problems 4–7 Dependency Structure

Problem	Main Topic	Key Tool
4	Elliptic Functions	Pole cancellation + Liouville on torus
5	Simply Connected Spaces	Contraction homotopy
6	Covering Spaces	Path lifting theorem
7	Complex Tori	Monodromy + Universal Cover

11 Concept Diagram





Modular Forms



Abelian Varieties



Modern Algebraic Geometry